

Course Name: Data Structure Using C & C++

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

Introduction to Data Structure and its Characteristics Array:- Representation of single and multidimensional arrays; Sparse arrays – lower and upper triangular matrices and Tri diagonal matrices with Vector Representation also.

UNIT-II

Stacks and Queues:- Introduction and primitive operations on stack; Stack application; Infix, postfix, prefix expressions; Evaluation of postfix expression; Conversion between prefix, infix and postfix, introduction and primitive operation on queues, D- queues and priority queues.

UNIT-III

Lists:- Introduction to linked lists; Sequential and linked lists, operations such as traversal, insertion, deletion searching, Two way lists and Use of headers.

UNIT-IV

Trees:- Introduction and terminology; Traversal of binary trees; Recursive algorithms for tree operations such as traversal, insertion, deletion; Binary Search Tree.

UNIT-V

B-Trees:- Introduction, The invention of B-Tree; Statement of the problem; Indexing with binary search trees; a better approach to tree indexes; B-Trees; working up from the bottom; Example for creating a B-Tree.

UNIT-V

Sorting Techniques:- Insertion sort, selection sort, merge sort, heap sort, searching Techniques: linear search, binary search and hashing.